Fundamental Limits of Reconstruction from Repeated DP Aggregates: A Cramér–Rao Perspective

Chenyue Zhang, *Member, IEEE*, Andrew Campbell, *Student Member, IEEE*, Anna Scaglione, *Fellow, IEEE*, Sean Peisert, *Senior Member, IEEE*

***Abstract*—We investigate the fundamental limits of value reconstruction from repeated differentially private aggregate queries. Modeling reconstruction as a parameter estimation problem with nuisance parameters, we derive a Cramér–Rao Bound (CRB) that quantifies the minimum Mean Squared Error (MSE) achievable by any unbiased adversary. Structurally, we leverage the Deep Sets theorem to prove that for permutation-invariant aggregates, *membership variation* is a necessary condition for identifiability. We further analyze the canonical differencing attack, demonstrating that the Best Linear Unbiased Estimator (BLUE) achieves the derived CRB. Our results provide a closed-form mapping from privacy budgets (ε, δ) to reconstruction risk, offering a rigorous estimation-theoretic benchmark for assessing privacy leakage in repeated sensing and analytics.**

***Index Terms*—Differential privacy, Reconstruction attacks, Fisher information, Cramér–Rao bound, Gaussian noise**

I. Introduction

Differential privacy (DP) provides a principled notion of indistinguishability between neighboring datasets, quantifying privacy loss via the privacy budget (ε, δ) [1]–[3]. This abstraction has become the gold standard for privacy accounting and mechanism design [4]–[6]. Yet, despite its theoretical clarity, the operational meaning of a given (ε, δ)—particularly how it constrains an adversary's ability to *reconstruct* sensitive individual values—remains poorly understood [7]. This ambiguity is especially pressing in systems requiring *repeated releases* of aggregate statistics, such as federated learning [8], interactive dashboards, and periodic reporting [9], where privacy leakage inevitably accumulates over time.

While prior studies have extensively examined *membership inference* (determining if a record is in the dataset) [10]–[12], this binary risk does not capture the full spectrum of harm. For high-stakes data modalities like mobility traces, smart-meter readings, and physiological signals [13], [14], the dominant confidentiality risk is high-fidelity *value recovery*. Crucially, recent findings demonstrate that mechanisms secure against membership tests may still allow accurate numerical reconstruction [15]–[19]. This motivates a fundamental question: *Given a sequence of DP-protected aggregate releases, under what conditions does the accumulated information become sufficient to estimate a specific target record?*

Our starting point is a geometric observation about aggregate query interfaces. Most practical DP releases—means, sums, losses, histograms—are *permutation-invariant* statistics [9]. Using the Deep Sets decomposition [20], we show that permutation invariance imposes a rigid Jacobian footprint: for a fixed query form, repeated releases only rescale the same footprint and need not create new identifiable directions after eliminating unknown nuisance records. As a result, *membership design* becomes the key structural lever for identifiability.

Building on this geometry, we develop an estimation-theoretic framework that not only diagnoses when reconstruction is possible, but also yields *computable* attacks and *closed-form* budget-to-risk laws, where the risk is quantified by reconstruction MSE. We model reconstruction as estimation with nuisance parameters and characterize feasibility via the effective Fisher information. For linear aggregate workloads, the resulting Cramér–Rao limits are attainable: the optimal risk is achieved by an explicit best linear unbiased estimator, recovering classical differencing attacks as special cases and providing sharp MSE scalings in terms of (ε, δ) and the number of releases L.

II. Estimation-Theoretic Framework

This section formalizes the reconstruction attack as a statistical parameter estimation problem. We establish the observation model, incorporate differential privacy as a constraint on the noise added to query answers, and derive fundamental limits on reconstruction accuracy via the Cramér–Rao bound in the presence of nuisance parameters.

A. Observation Model and Threat Scenario

We consider a fixed dataset consisting of N records $\boldsymbol{x}_{1:N} := (\boldsymbol{x}_1, \dots, \boldsymbol{x}_N)$, where each $\boldsymbol{x}_k \in \mathbb{R}^d$. We adopt a worst-case threat model in which the records are treated as deterministic but unknown constants. The adversary's objective is to reconstruct a specific target record $\boldsymbol{x}_i$ solely from the mechanism's released outputs.

The adversary observes a sequence of L released query answers. Stacking all releases, we model the observation vector $\boldsymbol{y} \in \mathbb{R}^{Ld_y}$ as

$$\boldsymbol{y} = \boldsymbol{q}(\boldsymbol{x}_{1:N}) + \boldsymbol{w}, \qquad \boldsymbol{w} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Omega}), \quad \boldsymbol{\Omega} \succ \boldsymbol{0}, \tag{1}$$

where $\boldsymbol{q} : \mathbb{R}^{Nd} \to \mathbb{R}^{Ld_y}$ is the known deterministic query map, and $\boldsymbol{\Omega}$ captures the covariance of additive Gaussian

This work was supported by the Director, Cybersecurity, Energy Security, and Emergency Response (CESER) Office of U.S. Department of Energy via the Privacy-Preserving, Collective Cyberattack Defense of DERs Project DEAC02-05CH11231.

Chenyue Zhang, Andrew Campbell and Anna Scaglione are with Department of Electrical and Computer Engineering, Cornell University, Cornell Tech, New York, 10044 NY, USA (ac2458@cornell.edu, cz563@cornell.edu, as337@cornell.edu)

Sean Peisert is with the Lawrence Berkeley National Laboratory, CA USA (email: sppeisert@lbl.gov)

perturbations (e.g., the Gaussian mechanism under a chosen accounting rule).

To model reconstruction of a target record $\boldsymbol{x}_i$ in the presence of unknown data, we partition the parameter as $\boldsymbol{\theta} := (\boldsymbol{x}_i, \boldsymbol{x}_{-i})$, where $\boldsymbol{x}_{-i}$ concatenates all records excluding the target. With a slight abuse of notation, we write $\boldsymbol{q}(\boldsymbol{\theta}) := \boldsymbol{q}(\boldsymbol{x}_{1:N})$ under this identification. We quantify reconstruction risk by the MSE of an estimator $\hat{\boldsymbol{x}}_i(\boldsymbol{y})$; among locally unbiased estimators (at $\boldsymbol{x}_i$),

$$\mathrm{MSE}^*(\boldsymbol{x}_i) := \inf_{\hat{\boldsymbol{x}}_i} \mathbb{E}_{\boldsymbol{w}}\left[\|\hat{\boldsymbol{x}}_i(\boldsymbol{y}) - \boldsymbol{x}_i\|_2^2\right], \tag{2}$$

where the expectation is over the privacy noise $\boldsymbol{w}$.

B. Differential Privacy and Noise Calibration

Differential privacy constrains the distinguishability of neighboring datasets. We say $\mathcal{D}, \mathcal{D}' \in \mathcal{X}^N$ are adjacent, written $\mathcal{D} \sim \mathcal{D}'$, if they differ in exactly one record.

Definition 1 ((ε, δ)-Differential Privacy [1])**.** *A randomized mechanism $\mathcal{M} : \mathcal{X}^N \to \mathcal{Y}$ satisfies (ε, δ)-differential privacy if for all adjacent datasets $\mathcal{D} \sim \mathcal{D}'$ and all measurable sets $\mathcal{S} \subseteq \mathcal{Y}$,*

$$\Pr[\mathcal{M}(\mathcal{D}) \in \mathcal{S}] \le e^{\varepsilon} \Pr[\mathcal{M}(\mathcal{D}') \in \mathcal{S}] + \delta. \tag{3}$$

In this work we focus on Gaussian-perturbed releases of (vector-valued) aggregate statistics, as in (1). For example, under *independent per-release* Gaussian noise, $\boldsymbol{\Omega} = \mathrm{blkdiag}(\sigma_1^2 \mathbf{I}, \ldots, \sigma_L^2 \mathbf{I})$, where a common calibration rule sets [2], [4]

$$\sigma_\ell^2 = 2\,\Delta_{2,\ell}^2 \ln(1.25/\delta_\ell)/\epsilon_\ell^2. \tag{4}$$

with $\Delta_{2,\ell} := \sup_{\mathcal{D}\sim\mathcal{D}'} \|q_\ell(\mathcal{D}) - q_\ell(\mathcal{D}')\|_2$ denoting the ℓ_2-sensitivity of the ℓ-th release. Throughout, we treat $\boldsymbol{\Omega}$ as the object through which the DP budget and accounting rule enter the reconstruction problem.

C. Fisher Information and Nuisance-Aware CRB

We now relate the query map and the DP-induced noise to fundamental limits of reconstruction. Under the Gaussian model (1), the Fisher information matrix (FIM) with respect to the full parameter $\boldsymbol{\theta}$ is [21]

$$\boldsymbol{I}(\boldsymbol{\theta}) = \boldsymbol{J}(\boldsymbol{\theta})^\top \boldsymbol{\Omega}^{-1} \boldsymbol{J}(\boldsymbol{\theta}), \tag{5}$$

where $\boldsymbol{J}(\boldsymbol{\theta}) := \nabla_{\boldsymbol{\theta}} \boldsymbol{q}(\boldsymbol{\theta})$ is the Jacobian of the query map.

To isolate the information available about the target record $\boldsymbol{x}_i$, we partition the FIM as

$$\boldsymbol{I}(\boldsymbol{\theta}) = \begin{bmatrix} \boldsymbol{I}_{ii} & \boldsymbol{I}_{i,-i} \\ \boldsymbol{I}_{-i,i} & \boldsymbol{I}_{-i,-i} \end{bmatrix}. \tag{6}$$

The *effective* Fisher information for $\boldsymbol{x}_i$, accounting for unknown nuisance records $\boldsymbol{x}_{-i}$, is defined via the Schur complement:

$$\boldsymbol{I}_{\mathrm{eff}}(i) := \boldsymbol{I}_{ii} - \boldsymbol{I}_{i,-i} \boldsymbol{I}_{-i,-i}^{\dagger} \boldsymbol{I}_{-i,i}, \tag{7}$$

where $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse [22]. We use $(\cdot)^\dagger$ since rank deficiency can arise from the query Jacobian even when $\boldsymbol{\Omega} \succ \mathbf{0}$.

The Cramér–Rao bound [21] states that for any locally unbiased estimator $\hat{\boldsymbol{x}}_i$, the MSE satisfies

$$\mathrm{MSE}(\hat{\boldsymbol{x}}_i) \ge \mathrm{tr}\big(\boldsymbol{I}_{\mathrm{eff}}(i)^\dagger\big). \tag{8}$$

This bound is our primary reconstruction-risk metric: singular $\boldsymbol{I}_{\mathrm{eff}}(i)$ implies structural non-identifiability of $\boldsymbol{x}_i$, meaning its footprint lies in the nuisance span and cannot be separated by repeated noisy releases.

III. Structural Analysis: Deep Sets and Identifiability

In Section II, we established an estimation-theoretic diagnostic: finite-MSE reconstruction of a target record $\boldsymbol{x}_i$ requires the nuisance-eliminated information $\boldsymbol{I}_{\mathrm{eff}}(i)$ to be nonsingular. We now apply this diagnostic to *aggregate query* interfaces common in DP deployments (e.g., sums, averages, and histograms). A defining characteristic of such workloads is *permutation invariance*: for each release ℓ, $q_\ell(\boldsymbol{x})$ depends on the multiset $\{\boldsymbol{x}_k : k \in C_\ell\}$, and is unchanged by any permutation of the records within C_ℓ. In this section, we characterize the algebraic structure of these queries and determine when it allows for feasible reconstruction.

A. The geometry of permutation invariance

To analyze the sensitivity of invariant aggregates, we leverage the Deep Sets theorem [20], which states that any permutation-invariant function on finite sets admits the representation

$$q_\ell(\boldsymbol{x}) = \rho_\ell\left(\sum_{k\in C_\ell} \phi_\ell(\boldsymbol{x}_k)\right), \qquad s_\ell := \sum_{k\in C_\ell} \phi_\ell(\boldsymbol{x}_k), \tag{9}$$

where $C_\ell \subseteq \{1, \ldots, N\}$ is the membership set, $\phi_\ell : \mathbb{R}^d \to \mathbb{R}^{d_e}$ is a per-record embedding, and $\rho_\ell : \mathbb{R}^{d_e} \to \mathbb{R}^{d_y}$ is a post-aggregation map. Applying the chain rule, the Jacobian with respect to a target record $\boldsymbol{x}_i$ factors as

$$\nabla_{\boldsymbol{x}_i} q_\ell(\boldsymbol{x}) \;=\; \mathbb{I}(i \in C_\ell)\, \mathbf{J}_{\rho_\ell}(s_\ell)\, \mathbf{J}_{\phi_\ell}(\boldsymbol{x}_i), \tag{10}$$

For a fixed subset C_ℓ, all participating records share the same *aggregation Jacobian* $\mathbf{J}_{\rho_\ell}(s_\ell)$; record-specific variation enters only through $\mathbf{J}_{\phi_\ell}(\boldsymbol{x}_i)$.

B. Identifiability under a static inner map

We focus on a static-query setting with $\phi_\ell \equiv \phi$ across releases, which captures common deployments with fixed aggregation semantics [8], [9]. Extensions to different feature maps $\{\phi_\ell\}$ are discussed in the extended version. Under this assumption, (9) reduces to

$$q_\ell(\boldsymbol{x}) = \rho_\ell\left(\sum_{k\in C_\ell} \phi(\boldsymbol{x}_k)\right), \qquad s_\ell := \sum_{k\in C_\ell} \phi(\boldsymbol{x}_k). \tag{11}$$

For example, subset averages correspond to $\phi(x) = x$ with $\rho_\ell(s) = s/|C_\ell|$, yielding the empirical mean. Histogram queries over B bins are captured by the same permutation-invariant form by taking a one-hot embedding $\phi(x) \in \{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_B\} \subset \{0,1\}^B$, with ρ_ℓ returning bin counts.

We now characterize when a target record $\boldsymbol{x}_i$ can be separated from unknown nuisance records under repeated releases. Let $\boldsymbol{q}(\boldsymbol{x}) := (q_1(\boldsymbol{x}), \ldots, q_L(\boldsymbol{x}))$ denote the stacked noiseless queries, and let $\boldsymbol{J} := \nabla_{\boldsymbol{x}} \boldsymbol{q}(\boldsymbol{x})$ be the Jacobian with respect to all records. Let $\boldsymbol{\Omega} \succ \mathbf{0}$ be the noise covariance of the stacked releases and define the whitened Jacobian $\widetilde{\boldsymbol{J}} := \boldsymbol{\Omega}^{-1/2}\boldsymbol{J}$, partitioned as $\widetilde{\boldsymbol{J}} = [\widetilde{\boldsymbol{J}}_i \ \widetilde{\boldsymbol{J}}_{-i}]$.

Lemma 1. *Let $\boldsymbol{G}_{1:L} := \mathrm{col}\big(\mathbf{J}_{\rho_1}(s_1), \ldots, \mathbf{J}_{\rho_L}(s_L)\big)$ and $\widetilde{\boldsymbol{G}}_{1:L} := \boldsymbol{\Omega}^{-1/2}\boldsymbol{G}_{1:L}$. If $C_\ell \equiv C$ for all ℓ, then for any record k,*

$$\widetilde{\boldsymbol{J}}_k = \mathbb{I}(k \in C)\ \widetilde{\boldsymbol{G}}_{1:L}\,\mathbf{J}_\phi(\boldsymbol{x}_k), \tag{12}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function and hence $\mathrm{range}(\widetilde{\boldsymbol{J}}) = \mathrm{range}\Big(\widetilde{\boldsymbol{G}}_{1:L}\,\mathbf{J}_{\phi,C}\Big)$, where $\mathbf{J}_{\phi,C} := \mathrm{col}\big(\mathbf{J}_\phi(\boldsymbol{x}_k)\big)_{k\in C}$ stacks the blocks $\{\mathbf{J}_\phi(\boldsymbol{x}_k)\}_{k\in C}$ column-wise.

Proof. Since $C_\ell \equiv C$, for any ℓ and k the Jacobian factorization gives $\nabla_{\boldsymbol{x}_k} q_\ell(\boldsymbol{x}) = \mathbb{I}(k \in C)\,\mathbf{J}_{\rho_\ell}(s_\ell)\mathbf{J}_\phi(\boldsymbol{x}_k)$. Stacking over $\ell = 1, \ldots, L$ yields $\boldsymbol{J}_k = \mathbb{I}(k \in C)\,\boldsymbol{G}_{1:L}\mathbf{J}_\phi(\boldsymbol{x}_k)$, hence $\widetilde{\boldsymbol{J}}_k = \boldsymbol{\Omega}^{-1/2}\boldsymbol{J}_k = \mathbb{I}(k \in C)\,\widetilde{\boldsymbol{G}}_{1:L}\mathbf{J}_\phi(\boldsymbol{x}_k)$, which proves (12). Moreover, columns with $k \notin C$ are zero, while columns with $k \in C$ equal $\widetilde{\boldsymbol{G}}_{1:L}\mathbf{J}_\phi(\boldsymbol{x}_k)$, so $\mathrm{range}(\widetilde{\boldsymbol{J}}) = \mathrm{range}\big(\widetilde{\boldsymbol{G}}_{1:L}\mathbf{J}_{\phi,C}\big)$. □

Lemma 1 shows that repeated releases with $C_\ell \equiv C$ cannot introduce new directions beyond those spanned by the nuisance records after whitening. We now translate this geometric constraint into an estimation-theoretic consequence via the effective Fisher information.

Corollary 1. *If $\mathrm{range}(\widetilde{\boldsymbol{J}}_i) \subseteq \mathrm{range}(\widetilde{\boldsymbol{J}}_{-i})$, then the effective Fisher information vanishes, $\boldsymbol{I}_{\mathrm{eff}}(i) = \mathbf{0}$, and the Cramér–Rao bound diverges.*

Proof. Recall that under the Gaussian model, the FIM admits the Gram form $\boldsymbol{I}(\boldsymbol{\theta}) = \widetilde{\boldsymbol{J}}^\top\widetilde{\boldsymbol{J}}$ with $\widetilde{\boldsymbol{J}} := \boldsymbol{\Omega}^{-1/2}\boldsymbol{J}$, and write $\widetilde{\boldsymbol{J}} = [\widetilde{\boldsymbol{J}}_i \ \widetilde{\boldsymbol{J}}_{-i}]$. By the generalized Schur-complement / projection identity for Gram matrices (with Moore–Penrose inverses), we have

$$\boldsymbol{I}_{\mathrm{eff}}(i) = \widetilde{\boldsymbol{J}}_i^\top(\mathbf{I} - \mathbf{P}_{-i})\widetilde{\boldsymbol{J}}_i, \qquad \mathbf{P}_{-i} := \widetilde{\boldsymbol{J}}_{-i}\widetilde{\boldsymbol{J}}_{-i}^\dagger,$$

where $\mathbf{P}_{-i}$ is the orthogonal projector onto $\mathrm{range}(\widetilde{\boldsymbol{J}}_{-i})$; see, e.g., [22, Sec. 3.4]. If $\mathrm{range}(\widetilde{\boldsymbol{J}}_i) \subseteq \mathrm{range}(\widetilde{\boldsymbol{J}}_{-i})$ holds, then $\mathbf{P}_{-i}\widetilde{\boldsymbol{J}}_i = \widetilde{\boldsymbol{J}}_i$, and thus $\boldsymbol{I}_{\mathrm{eff}}(i) = \mathbf{0}$. □

Under static membership $C_\ell \equiv C$, Lemma 1 implies that repetition cannot change $\mathrm{range}(\widetilde{\boldsymbol{J}})$. This does *not* mean repetition is ill-posed; rather, with x_{-i} unknown, the target is non-identifiable whenever $\mathrm{range}(\widetilde{\boldsymbol{J}}_i) \subseteq \mathrm{range}(\widetilde{\boldsymbol{J}}_{-i})$, hence $\boldsymbol{I}_{\mathrm{eff}}(i) = 0$ and additional releases cannot help. Therefore, membership must vary across releases; in particular, a necessary condition is that there exist $\ell_1 \neq \ell_2$ such that

$$\mathbb{I}(i \in C_{\ell_1}) \neq \mathbb{I}(i \in C_{\ell_2}). \tag{13}$$

IV. Linear Queries and Optimal Reconstruction

We assume the adversary knows the query interface, the membership sets $\{C_\ell\}_{\ell=1}^L$, and the noise model (equivalently, the DP-calibrated covariance $\boldsymbol{\Omega}$), but has no side information about the non-target records x_{-i}. As established in Section III, identifiability hinges on membership variation. We now specialize to linear subset queries, yielding a linear–Gaussian model in which the optimal reconstruction risk admits a closed form and is achieved by the BLUE. Due to space constraints, we relegate the missing proofs from this section to the supplementary materials.

A. Linear–Gaussian Instantiation

We specialize to the scalar case $d = 1$ and write $\boldsymbol{x}_i = x_i$ for simplicity; the vector-valued extension follows by Kronecker lifting (see appendix). We consider linear aggregate queries (e.g., subset sums or averages). In the Deep Sets representation, this corresponds to the identity feature map $\phi(x) = x$. Stacking L releases yields the linear observation model

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{w}, \qquad \boldsymbol{w} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}), \tag{14}$$

where $\boldsymbol{x} := (x_1, \ldots, x_N)^\top \in \mathbb{R}^N$, $\boldsymbol{y} \in \mathbb{R}^L$, and $\boldsymbol{H} \in \mathbb{R}^{L\times N}$ is the design matrix induced by the membership pattern. For subset averages, $H_{\ell i} = \frac{1}{|C_\ell|}\,\mathbb{I}(i \in C_\ell)$.

This linear model is a direct specialization of the general nuisance-aware criterion in Section II: since $\boldsymbol{q}(\boldsymbol{x}) = \boldsymbol{H}\boldsymbol{x}$ in (14), the Jacobian equals the design matrix, $\boldsymbol{J} = \boldsymbol{H}$, and the whitened geometry is governed by $\widetilde{\boldsymbol{H}} := \boldsymbol{\Omega}^{-1/2}\boldsymbol{H}$. Partition $\widetilde{\boldsymbol{H}} = [\widetilde{\boldsymbol{h}}_i \ \widetilde{\boldsymbol{H}}_{-i}]$. Then finite-MSE reconstruction of x_i in the presence of nuisance parameters $\boldsymbol{x}_{-i}$ is possible iff

$$\widetilde{\boldsymbol{h}}_i \notin \mathrm{range}\big(\widetilde{\boldsymbol{H}}_{-i}\big). \tag{15}$$

B. Attainability of the Limit: Optimality of BLUE

Under (14), the Fisher information matrix is $\boldsymbol{I} = \boldsymbol{H}^\top\boldsymbol{\Omega}^{-1}\boldsymbol{H}$. When (15) holds, the coordinate-wise Cramér–Rao bound for estimating x_i in the presence of nuisance parameters $\boldsymbol{x}_{-i}$ can be written as

$$\mathrm{MSE}^*(x_i) = \boldsymbol{e}_i^\top \left(\boldsymbol{H}^\top\boldsymbol{\Omega}^{-1}\boldsymbol{H}\right)^\dagger \boldsymbol{e}_i, \tag{16}$$

A defining feature of the linear–Gaussian model is that this lower bound is *tight*. The BLUE minimizes the variance among all unbiased estimators and achieves the risk in (16).

Proposition 1 (Optimality of BLUE)**.** *Let $\hat{x}_i = \boldsymbol{v}^\top\boldsymbol{y}$ be a linear estimator. The solution to*

$$\min_{\boldsymbol{v}\in\mathbb{R}^L} \ \boldsymbol{v}^\top\boldsymbol{\Omega}\boldsymbol{v} \quad \text{s.t.} \quad \boldsymbol{v}^\top\boldsymbol{H} = \boldsymbol{e}_i^\top \tag{17}$$

is the BLUE [21], and its variance satisfies

$$\mathrm{Var}(\hat{x}_{i,\mathrm{BLUE}}) = \mathrm{MSE}^*(x_i). \tag{18}$$

C. Case Study: Optimal Differencing via Repeated IN/OUT

We instantiate the linear–Gaussian model (14) for the canonical differencing attack on subset averages [23]–[26]. Without loss of generality, fix the target index $i = 1$ and choose a background membership set $C_{\mathrm{out}} \subseteq \{2, \ldots, N\}$ with $|C_{\mathrm{out}}| = n$. Define the target-inclusive set

$$C_{\mathrm{in}} := \{1\} \cup C_{\mathrm{out}}, \qquad |C_{\mathrm{in}}| = n + 1.$$

Across L releases, the adversary repeats two subset-average queries with these memberships: IN releases use $C_\ell = C_{\text{in}}$, while OUT releases use $C_\ell = C_{\text{out}}$. Let L_{in} and L_{out} denote the numbers of IN and OUT releases, respectively, with $L = L_{\text{in}} + L_{\text{out}}$. Let $\bar{y}_{\text{in}}$ and $\bar{y}_{\text{out}}$ denote the empirical means of the released answers within the two blocks.

For subset averages, the noiseless answers are

$$q_{\text{in}}(\boldsymbol{x}) = \frac{1}{|C_{\text{in}}|} \sum_{k \in C_{\text{in}}} x_k = \frac{1}{n+1}\Big(x_1 + \sum_{k \in C_{\text{out}}} x_k\Big),$$
$$q_{\text{out}}(\boldsymbol{x}) = \frac{1}{|C_{\text{out}}|} \sum_{k \in C_{\text{out}}} x_k = \frac{1}{n} \sum_{k \in C_{\text{out}}} x_k. \tag{19}$$

Assume independent Gaussian perturbations with per-release variances σ_{in}^2 and σ_{out}^2 for the IN and OUT releases, respectively. At this stage, $(\sigma_{\text{in}}^2, \sigma_{\text{out}}^2)$ are treated as generic noise parameters (not necessarily DP-calibrated); we relate them to (ε, δ) via Gaussian-mechanism calibration and a specified accounting rule in Corollary 3. For the IN and OUT blocks, the block means satisfy

$$\bar{y}_{\text{in}} = q_{\text{in}}(\boldsymbol{x}) + \bar{w}_{\text{in}}, \qquad \bar{y}_{\text{out}} = q_{\text{out}}(\boldsymbol{x}) + \bar{w}_{\text{out}},$$

with $\bar{w}_{\text{in}} \sim \mathcal{N}\big(0, \sigma_{\text{in}}^2/L_{\text{in}}\big)$ and $\bar{w}_{\text{out}} \sim \mathcal{N}\big(0, \sigma_{\text{out}}^2/L_{\text{out}}\big)$.

Corollary 2 (Optimal differencing estimator and allocation)**.** *For the repeated IN/OUT design above, the BLUE of x_1 is*

$$\hat{x}_1 \;=\; (n+1)\,\bar{y}_{\text{in}} \;-\; n\,\bar{y}_{\text{out}}. \tag{20}$$

Its minimum achievable mean squared error is

$$\mathrm{MSE}^*(x_1) = \frac{(n+1)^2\sigma_{\text{in}}^2}{L_{\text{in}}} + \frac{n^2\sigma_{\text{out}}^2}{L_{\text{out}}}.$$

For a fixed total number of releases $L = L_{\text{in}} + L_{\text{out}}$, this error is minimized when

$$\frac{L_{\text{in}}}{L_{\text{out}}} = \frac{n+1}{n}\,\frac{\sigma_{\text{in}}}{\sigma_{\text{out}}}, \tag{21}$$

in which case

$$\min_{L_{\text{in}}+L_{\text{out}}=L} \mathrm{MSE}^*(x_1) = \frac{1}{L}\Big((n+1)\sigma_{\text{in}} + n\sigma_{\text{out}}\Big)^2. \tag{22}$$

We now connect this reconstruction risk to the differential privacy budget. The following result reveals a fundamental scaling law: for subset-average queries, the sensitivity reduction from enlarging the background set is exactly offset by the noise required for reconstruction.

Corollary 3. *Suppose a total budget $(\varepsilon_{\text{tot}}, \delta_{\text{tot}})$ is split uniformly across L releases (Basic Composition [2], [6]), i.e., $\varepsilon_\ell = \varepsilon_{\text{tot}}/L$ and $\delta_\ell = \delta_{\text{tot}}/L$ for $\ell = 1, \dots, L$, and each release is calibrated using the Gaussian mechanism (4) under record-level adjacency radius Δ. For subset averages, $\Delta_{2,\ell} = \Delta/|C_\ell|$. Under the repeated IN/OUT design and the optimal allocation in Corollary 2,*

$$\min_{L_{\text{in}}+L_{\text{out}}=L} \mathrm{MSE}^*(x_1) = \frac{8L\,\ln\big(1.25L/\delta_{\text{tot}}\big)}{\varepsilon_{\text{tot}}^2}\,\Delta^2. \tag{23}$$

Corollary 3 shows that, under a fixed total budget $(\varepsilon_{\text{tot}}, \delta_{\text{tot}})$ with uniform splitting, increasing the number of releases does not improve the attainable reconstruction accuracy for the repeated differencing attack. For subset averages, the apparent sensitivity gain from enlarging the background set ($\Delta_{2,\ell} \propto 1/|C_\ell|$) is exactly canceled by the BLUE differencing amplification, so the resulting risk is independent of n. More importantly, uniform splitting gives the variance scale per-release as $\sigma_\ell^2 \propto (L/\varepsilon_{\text{tot}})^2$ (up to logarithmic factors), while averaging only provides a reduction $1/L$, resulting in the monotone growth $\Theta(L \log L)$ in (23). Consequently, if the adversary can choose the query count under the same total budget, the optimal strategy is to **use the minimum informative releases $L = 2$ and allocate the entire $(\varepsilon_{\text{tot}}, \delta_{\text{tot}})$ budget to this single pair.**

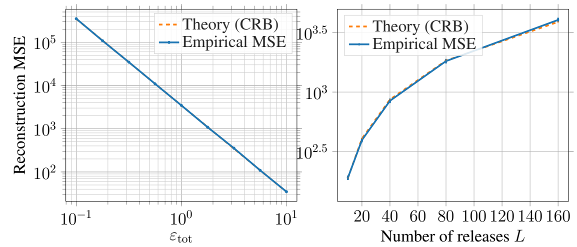

Fig. 1: Empirical reconstruction MSE of the BLUE versus the theoretical prediction. Left: sweep ε_{tot} with L fixed. Right: sweep L under fixed $(\varepsilon_{\text{tot}}, \delta_{\text{tot}})$.

V. Numerical Experiments

We evaluate the repeated IN/OUT subset-average design from Section IV in the scalar setting ($d = 1$) with target $i = 1$. Each release reports a subset average over $C_{\text{in}} = \{1\} \cup C_{\text{out}}$ and C_{out}, perturbed by independent Gaussian noise calibrated by (4) under uniform budget splitting. We estimate the reconstruction MSE of the BLUE (Corollary 2) via Monte Carlo and compare it to the closed-form prediction (23).

Figure 1 (Left) shows that the empirical MSE essentially coincides with the theoretical curve across ε_{tot}, confirming that the CRB is attained by the BLUE in this linear–Gaussian setting. Figure 1 (Right) shows that, under a fixed total budget, the MSE increases with the number of releases L, matching the predicted $L \log L$ scaling in (23).

VI. Conclusion

This work provides an estimation-theoretic benchmark for reconstruction from repeated DP aggregate releases by using a nuisance-aware Cramér–Rao bound to characterize fundamental MSE limits. Structurally, using the Deep Sets representation, we show that for permutation-invariant statistics, membership variation across releases is necessary for identifiability in the presence of unknown nuisance records. Quantitatively, in the linear–Gaussian regime, we prove CRB tightness by showing that the BLUE attains the bound and yields closed-form budget-to-risk scaling laws: under uniform splitting of a fixed total budget, the minimum achievable reconstruction MSE scales as $\Theta(L \log L)$, so increasing the number of releases strictly worsens the adversary's achievable reconstruction accuracy. We focus on static query setting, the extensions to time-varying feature maps ($\phi_\ell \neq \phi_{\ell'}$) are left for future work.

References

[1] C. Dwork, F. McSherry, K. Nissim, and A. Smith, "Calibrating noise to sensitivity in private data analysis," in *Theory of Cryptography*, ser. Lecture Notes in Computer Science, vol. 3876. Springer, 2006, pp. 265–284.

[2] C. Dwork and A. Roth, *The Algorithmic Foundations of Differential Privacy*, ser. Foundations and Trends in Theoretical Computer Science. Hanover, MA, USA: Now Publishers, 2014, vol. 9, no. 3–4.

[3] J. Dong, A. Roth, and W. J. Su, "Gaussian differential privacy," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 84, no. 1, pp. 3–37, 2022.

[4] B. Balle and Y.-X. Wang, "Improving the Gaussian mechanism for differential privacy: Analytical calibration and optimal denoising," in *Proceedings of the 35th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 80. PMLR, 2018, pp. 394–403.

[5] I. Mironov, "Rényi differential privacy," *2017 IEEE 30th Computer Security Foundations Symposium (CSF)*, 2017.

[6] P. Kairouz, S. Oh, and P. Viswanath, "The composition theorem for differential privacy," *IEEE Transactions on Information Theory*, vol. 63, no. 6, pp. 4037–4049, 2017.

[7] L. Wasserman and S. Zhou, "A statistical framework for differential privacy," *Journal of the American Statistical Association*, vol. 105, no. 489, pp. 375–389, 2010.

[8] H. B. McMahan, D. Ramage, K. Talwar, and L. Zhang, "Learning differentially private recurrent language models," in *International Conference on Learning Representations (ICLR)*, 2018.

[9] C. Dwork, M. Naor, T. Pitassi, and G. N. Rothblum, "Differential privacy under continual observation," in *Proceedings of the forty-second ACM symposium on Theory of computing*, 2010, pp. 715–724.

[10] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," in *Proceedings of the IEEE Symposium on Security and Privacy (S&P)*, 2017, pp. 3–18.

[11] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, "Privacy risk in machine learning: Analyzing the connection to overfitting," in *Proceedings of the IEEE Computer Security Foundations Symposium (CSF)*, 2018, pp. 268–282.

[12] A. Salem, Y. Zhang, M. Humbert, P. Berrang, and M. Fritz, "Ml-leaks: Model and data independent membership inference attacks and defenses on machine learning models," in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2019.

[13] Y.-A. de Montjoye, C. A. Hidalgo, M. Verleysen, and V. D. Blondel, "Unique in the crowd: The privacy bounds of human mobility," *Scientific Reports*, vol. 3, p. 1376, 2013.

[14] N. Apthorpe, D. Reisman, and N. Feamster, "A smart home is no castle: Privacy vulnerabilities of encrypted iot traffic," in *Workshop on Data and Algorithmic Transparency (DAT)*, 2017.

[15] F. Tramer and D. Boneh, "Differentially private learning needs better features (or much more data)," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.

[16] C. Guo, B. Karrer, K. Chaudhuri, and L. van der Maaten, "Bounding training data reconstruction in private (deep) learning," in *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022, pp. 8056–8071.

[17] L. Zhu, Z. Liu, and S. Han, "Deep leakage from gradients," *Advances in neural information processing systems*, vol. 32, 2019.

[18] J. Geiping, H. Bauermeister, H. Dröge, and M. Moeller, "Inverting gradients-how easy is it to break privacy in federated learning?" *Advances in neural information processing systems*, vol. 33, pp. 16 937–16 947, 2020.

[19] J. Hayes, B. Balle, and S. Mahloujifar, "Bounding training data reconstruction in dp-sgd," *Advances in neural information processing systems*, vol. 36, pp. 78 696–78 722, 2023.

[20] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. Salakhutdinov, and A. J. Smola, "Deep sets," in *Advances in Neural Information Processing Systems*, 2017, pp. 3391–3401.

[21] S. M. Kay, *Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory*. Prentice Hall, 1993.

[22] A. Ben-Israel and T. N. E. Greville, *Generalized Inverses: Theory and Applications*, 2nd ed. Springer, 2003.

[23] I. Dinur and K. Nissim, "Revealing information while preserving privacy," in *Proceedings of the 22nd ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems (PODS)*. ACM, 2003, pp. 202–210.

[24] F. McSherry and K. Talwar, "Mechanism design via differential privacy," in *48th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*. IEEE, 2007, pp. 94–103.

[25] A. Cohen and K. Nissim, "Linear program reconstruction in practice," *Journal of Privacy and Confidentiality*, vol. 10, no. 1, 2020.

[26] C. Dwork, F. McSherry, and K. Talwar, "The price of privacy and the limits of lp decoding," in *Proceedings of the thirty-ninth annual ACM symposium on Theory of computing*, 2007, pp. 85–94.

Supplementary Material

Appendix A
Vector-valued extension via Kronecker lifting

Let $\boldsymbol{x}_k \in \mathbb{R}^d$ and $\boldsymbol{x}_i := \vec{(\boldsymbol{X})} \in \mathbb{R}^{Nd}$ with $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_N]^\top$. For coordinate-wise linear aggregates (e.g., subset sums/averages), stacking L releases gives

$$\boldsymbol{y} = (\boldsymbol{H} \otimes \boldsymbol{I}_d)\boldsymbol{x}_i + \boldsymbol{w}, \qquad \boldsymbol{w} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Omega} \otimes \boldsymbol{I}_d), \tag{24}$$

where $\boldsymbol{H} \in \mathbb{R}^{L\times N}$ is the scalar membership design. The Fisher information is

$$\boldsymbol{I}_{\boldsymbol{x}_i} = (\boldsymbol{H} \otimes \boldsymbol{I}_d)^\top (\boldsymbol{\Omega} \otimes \boldsymbol{I}_d)^{-1} (\boldsymbol{H} \otimes \boldsymbol{I}_d) = (\boldsymbol{H}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{H}) \otimes \boldsymbol{I}_d. \tag{25}$$

For target record $\boldsymbol{x}_i \in \mathbb{R}^d$ (others treated as nuisance), the effective information for the block $\boldsymbol{x}_i$ inherits the same lifting:

$$\boldsymbol{I}_{\text{eff}}(i) = \boldsymbol{I}^{(1)}_{\text{eff}}(i) \otimes \boldsymbol{I}_d, \qquad \boldsymbol{I}^{(1)}_{\text{eff}}(i) \text{ as in the scalar model.}$$

Hence the nuisance-aware CRB implies

$$\inf_{\hat{\boldsymbol{x}}_i \text{ unbiased}} \mathbb{E}\|\hat{\boldsymbol{x}}_i - \boldsymbol{x}_i\|_2^2 = \operatorname{tr}\Big(\boldsymbol{I}_{\text{eff}}(i)^\dagger\Big) = d \cdot \boldsymbol{e}_i^\top (\boldsymbol{H}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{H})^\dagger \boldsymbol{e}_i, \tag{26}$$

whenever the scalar identifiability condition (15) holds.

Appendix B
Proof of Proposition 1

Consider the linear–Gaussian model (14): $\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{w}$, $\boldsymbol{w} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Omega})$ with $\boldsymbol{\Omega} \succ \boldsymbol{0}$. Fix a target coordinate x_i and a linear estimator $\hat{x}_i = \boldsymbol{v}^\top \boldsymbol{y}$. Unbiasedness for all $\boldsymbol{x}$ requires

$$\mathbb{E}[\hat{x}_i] = \boldsymbol{v}^\top \boldsymbol{H}\boldsymbol{x} = x_i = \boldsymbol{e}_i^\top \boldsymbol{x} \quad \forall \boldsymbol{x}, \qquad \Longleftrightarrow \qquad \boldsymbol{v}^\top \boldsymbol{H} = \boldsymbol{e}_i^\top. \tag{27}$$

Under (27), the variance equals

$$\operatorname{Var}(\hat{x}_i) = \operatorname{Var}(\boldsymbol{v}^\top \boldsymbol{w}) = \boldsymbol{v}^\top \boldsymbol{\Omega} \boldsymbol{v}, \tag{28}$$

so the minimum-variance unbiased linear estimator is obtained by solving (17). Form the Lagrangian

$$\mathcal{L}(\boldsymbol{v}, \boldsymbol{\lambda}) = \boldsymbol{v}^\top \boldsymbol{\Omega} \boldsymbol{v} + \boldsymbol{\lambda}^\top (\boldsymbol{H}^\top \boldsymbol{v} - \boldsymbol{e}_i),$$

where we used $(\boldsymbol{v}^\top \boldsymbol{H} = \boldsymbol{e}_i^\top) \Leftrightarrow (\boldsymbol{H}^\top \boldsymbol{v} = \boldsymbol{e}_i)$. Stationarity gives

$$\nabla_{\boldsymbol{v}} \mathcal{L} = 2\boldsymbol{\Omega}\boldsymbol{v} + \boldsymbol{H}\boldsymbol{\lambda} = \boldsymbol{0} \quad \Longrightarrow \quad \boldsymbol{v} = -\tfrac{1}{2}\boldsymbol{\Omega}^{-1}\boldsymbol{H}\boldsymbol{\lambda}. \tag{29}$$

Enforcing $\boldsymbol{H}^\top \boldsymbol{v} = \boldsymbol{e}_i$ and letting $\boldsymbol{\gamma} := -(1/2)\boldsymbol{\lambda}$ yields

$$\boldsymbol{H}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{H}\, \boldsymbol{\gamma} = \boldsymbol{e}_i. \tag{30}$$

When $\boldsymbol{H}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{H}$ is singular, the minimum-variance solution is obtained with the Moore–Penrose pseudoinverse,

$$\boldsymbol{\gamma} = (\boldsymbol{H}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{H})^\dagger \boldsymbol{e}_i, \qquad \boldsymbol{v}_{\text{BLUE}} = \boldsymbol{\Omega}^{-1}\boldsymbol{H}(\boldsymbol{H}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{H})^\dagger \boldsymbol{e}_i. \tag{31}$$

Substituting (31) into (28) and using symmetry of $\boldsymbol{H}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{H}$ gives

$$\operatorname{Var}(\hat{x}_{i,\text{BLUE}}) = \boldsymbol{v}_{\text{BLUE}}^\top \boldsymbol{\Omega} \boldsymbol{v}_{\text{BLUE}} = \boldsymbol{e}_i^\top (\boldsymbol{H}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{H})^\dagger \boldsymbol{e}_i. \tag{32}$$

Finally, the coordinate-wise nuisance-aware CRB for estimating x_i in the linear–Gaussian model equals $\boldsymbol{e}_i^\top (\boldsymbol{H}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{H})^\dagger \boldsymbol{e}_i$ (cf. (16)), so (32) proves that the BLUE attains the bound, i.e., $\operatorname{Var}(\hat{x}_{i,\text{BLUE}}) = \text{MSE}^*(x_i)$.

Appendix C
Proof of Corollary 2

Fix $i = 1$ and let $C_{\text{out}} \subseteq \{2, \dots, N\}$ with $|C_{\text{out}}| = n$ and $C_{\text{in}} = \{1\} \cup C_{\text{out}}$. Define the nuisance sum $S := \sum_{k \in C_{\text{out}}} x_k$. For subset averages, the noiseless answers are

$$q_{\text{in}} = (x_1 + S)/(n+1), \qquad q_{\text{out}} = S/n.$$

Let $\bar{y}_{\text{in}}$ and $\bar{y}_{\text{out}}$ be the empirical means of the released answers in the IN and OUT blocks. Under independent Gaussian perturbations, we have

$$\bar{y}_{\text{in}} = q_{\text{in}} + \bar{w}_{\text{in}}, \qquad \bar{y}_{\text{out}} = q_{\text{out}} + \bar{w}_{\text{out}},$$

where $\bar{w}_{\text{in}} \sim \mathcal{N}(0, \sigma_{\text{in}}^2 / L_{\text{in}})$ and $\bar{w}_{\text{out}} \sim \mathcal{N}(0, \sigma_{\text{out}}^2 / L_{\text{out}})$, independent across blocks.

Consider any linear estimator $\hat{x}_1 = a\, \bar{y}_{\text{in}} + b\, \bar{y}_{\text{out}}$. Unbiasedness for all (x_1, S) requires

$$\mathbb{E}[\hat{x}_1] = a\frac{x_1 + S}{n+1} + b\frac{S}{n} = x_1 \quad \forall (x_1, S),$$

which implies the coefficient constraints

$$\frac{a}{n+1} = 1, \qquad \frac{a}{n+1} + \frac{b}{n} = 0.$$

Solving yields $a = n+1$ and $b = -n$, giving the differencing estimator

$$\hat{x}_1 = (n+1)\bar{y}_{\text{in}} - n\bar{y}_{\text{out}},$$

which is the unique unbiased linear estimator and hence is the BLUE.

Since the block noises are independent,

$$\text{MSE}(\hat{x}_1) = \operatorname{Var}(\hat{x}_1) = (n+1)^2 \frac{\sigma_{\text{in}}^2}{L_{\text{in}}} + n^2 \frac{\sigma_{\text{out}}^2}{L_{\text{out}}},$$

proving the stated MSE formula. To optimize the allocation under $L_{\text{in}} + L_{\text{out}} = L$, minimize $A/L_{\text{in}} + B/L_{\text{out}}$ with $A = (n+1)^2\sigma_{\text{in}}^2$ and $B = n^2\sigma_{\text{out}}^2$. The first-order condition gives

$$-\frac{A}{L_{\text{in}}^2} + \frac{B}{L_{\text{out}}^2} = 0 \quad \Longrightarrow \quad \frac{L_{\text{in}}}{L_{\text{out}}} = \sqrt{\frac{A}{B}} = \frac{n+1}{n}\frac{\sigma_{\text{in}}}{\sigma_{\text{out}}},$$

which is (21). At this ratio, the minimum value is

$$\min_{L_{\text{in}} + L_{\text{out}} = L} \text{MSE}^*(x_1) = \frac{(\sqrt{A} + \sqrt{B})^2}{L} = \frac{1}{L}\Big((n+1)\sigma_{\text{in}} + n\sigma_{\text{out}}\Big)^2,$$

proving (22).

Appendix D
Proof of Corollary 3

Under uniform splitting, $\varepsilon_\ell = \varepsilon_{\text{tot}}/L$ and $\delta_\ell = \delta_{\text{tot}}/L$. For subset averages, $\Delta_{2,\text{in}} = \Delta/(n+1)$ and $\Delta_{2,\text{out}} = \Delta/n$. By the Gaussian calibration (4),

$$\sigma_{\text{in}}^2 = \frac{\kappa\,\Delta^2}{(n+1)^2}, \qquad \sigma_{\text{out}}^2 = \frac{\kappa\,\Delta^2}{n^2}, \qquad \kappa := 2\ln\Big(\frac{1.25L}{\delta_{\text{tot}}}\Big)\Big(\frac{L}{\varepsilon_{\text{tot}}}\Big)^2.$$

Hence $(n+1)\sigma_{\text{in}} = n\sigma_{\text{out}}$, so the optimal split (21) gives $L_{\text{in}} = L_{\text{out}} = L/2$. Plugging into (22) yields

$$\text{MSE}^*(x_1) = \frac{1}{L}\Big((n+1)\sigma_{\text{in}} + n\sigma_{\text{out}}\Big)^2 = \frac{8L\ln(1.25L/\delta_{\text{tot}})}{\varepsilon_{\text{tot}}^2}\Delta^2,$$

which is independent of n.

Information for Reproducibility

We report the empirical mean-squared error (MSE) of the BLUE under the repeated IN/OUT subset-average design described in the main text. All simulations are performed in accordance with the model and noise calibration specified in Section V.

We fix the population size to $N = 2000$, which is sufficiently large to ensure $n \leq N - 1$ without boundary effects, and set the subset size to $n = 50$. Unless otherwise stated, we use $\Delta = 1$ and a total failure probability $\delta_{\mathrm{tot}} = 10^{-3}$. We consider two parameter sweeps: (i) a privacy sweep, where $\varepsilon_{\mathrm{tot}} \in \{0.1, \ldots, 10\}$ is logarithmically spaced while the number of releases is fixed to $L = 40$; and (ii) a release sweep, where $L \in \{10, 20, 40, 80, 160\}$ while the total privacy budget is fixed to $\varepsilon_{\mathrm{tot}} = 2.0$.

Under uniform budget allocation, each release uses $\varepsilon_\ell = \varepsilon_{\mathrm{tot}}/L$ and $\delta_\ell = \delta_{\mathrm{tot}}/L$. Each query answer is perturbed with independent Gaussian noise calibrated according to (4), using the appropriate ℓ_2-sensitivity $\Delta_{2,\ell} = \Delta/|C_\ell|$, where $|C_\ell| = n + 1$ for IN releases and $|C_\ell| = n$ for OUT releases.

For each configuration, we perform $T = 4000$ independent Monte Carlo trials. In each trial, we draw independent privacy noises for all L releases, form the corresponding noisy answers, and compute the BLUE estimate $\hat{x}_1$ using the closed-form expression in (20). The squared reconstruction error $(\hat{x}_1 - x_1)^2$ is recorded for each trial. The empirical MSE is then computed as the sample mean of the squared errors across the T trials.

To quantify statistical uncertainty, we report 95% confidence intervals for the empirical MSE using the normal approximation

$$\widehat{\mathrm{MSE}} \;\pm\; 1.96\,\frac{\widehat{s}}{\sqrt{T}},$$

where $\widehat{s}$ denotes the sample standard deviation of the squared errors. Theoretical predictions are given by the closed-form expression in (23), and are overlaid with the empirical results for comparison.